

WINDOW ROCK, AZ, USA

WMO: 722764

Lat: 35.658N

Lon: 109.061W

Elev: 2054

StdP: 78.96

Time Zone: -7.00 (NAM)

Period: 99-19

WBAN: 03029

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-14.4	-22.9	0.6	-6.4	-21.0	0.7	-4.0	11.2	5.8	9.7	4.3	0.4	190	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	32.5	13.5	31.2	13.3	30.0	13.0	16.6	24.6	16.0	24.2	15.5	23.7	4.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	13.3	17.8	13.8	12.7	17.7	12.9	11.9	17.8	54.1	24.4	52.2	24.0	50.4	23.3	19.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	9.4	8.1	DB	-22.6	34.6	3.5	1.1	-25.1	35.4	-27.2	36.1	-29.1	36.8	-31.7	37.6	(4)
(5)			WB	-22.9	17.9	3.4	0.9	-25.4	18.5	-27.3	19.1	-29.2	19.5	-31.7	20.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-1.5	0.7	4.6	8.2	12.6	18.5	21.3	19.7	15.8	9.5	3.5	-1.3	(6)	
	DBStd	8.66	4.49	4.22	3.58	3.40	3.63	2.85	1.81	1.82	2.85	3.53	3.94	4.36	(7)	
	HDD10.0	1474	357	259	169	72	17	0	0	0	2	51	195	352	(8)	
	HDD18.3	3468	615	493	426	304	178	31	1	8	83	274	444	610	(9)	
	CDD10.0	1239	0	0	2	18	99	256	352	301	176	35	1	0	(10)	
	CDD18.3	191	0	0	0	0	2	37	94	51	7	0	0	0	(11)	
	CDH23.3	4072	0	0	0	8	210	1247	1367	858	358	23	0	0	(12)	
	CDH26.7	1424	0	0	0	0	37	491	559	267	71	1	0	0	(13)	
(14) Wind	WSAvg	2.5	1.9	2.5	2.9	3.7	3.4	3.1	2.3	2.0	2.1	2.2	2.1	2.0	(14)	
(15) Precipitation	PrecAvg	252	19	20	18	13	12	6	33	42	28	23	16	21	(15)	
	PrecMax	365	86	95	55	36	73	23	96	91	85	73	42	55	(16)	
	PrecMin	121	3	2	0	2	0	1	3	3	7	0	0	2	(17)	
	PrecStd	59	18	19	14	10	16	6	23	23	21	20	14	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.7	17.8	22.2	24.6	29.9	34.3	34.1	32.5	30.5	26.0	21.0	14.9	(19)	
		MCWB	4.6	5.5	7.7	8.6	11.8	13.0	14.2	13.9	13.1	10.5	7.6	4.5	(20)	
	2%	DB	12.3	15.6	19.6	22.4	27.4	32.5	32.7	30.9	28.6	23.6	18.9	12.5	(21)	
		MCWB	3.5	4.7	6.5	7.6	10.6	12.6	14.1	13.8	12.4	9.5	6.5	3.5	(22)	
	5%	DB	10.0	12.5	17.6	21.0	25.9	31.1	31.4	29.5	27.2	22.1	16.6	10.1	(23)	
		MCWB	2.3	3.1	5.6	7.1	9.5	12.1	14.0	13.5	12.2	9.1	5.7	2.7	(24)	
	10%	DB	7.4	9.8	15.4	19.0	23.9	29.7	30.0	28.0	25.6	20.1	13.9	7.6	(25)	
		MCWB	1.4	2.2	4.5	6.2	8.9	11.5	14.0	13.9	11.9	8.3	4.5	1.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.9	6.9	8.3	9.4	12.9	15.3	17.4	17.3	15.7	13.2	8.5	6.3	(27)	
		MCDB	12.3	14.6	20.1	21.8	27.7	26.6	25.5	24.1	22.5	18.7	16.9	11.6	(28)	
	2%	WB	4.2	5.2	7.0	8.3	11.3	14.4	16.8	16.6	14.7	11.4	7.4	4.6	(29)	
		MCDB	10.3	13.0	18.0	20.7	24.7	26.7	25.2	24.0	22.6	19.5	16.1	10.2	(30)	
	5%	WB	3.0	4.0	6.0	7.4	10.4	13.6	16.2	16.0	14.0	10.3	6.3	3.3	(31)	
		MCDB	7.8	11.1	16.1	19.4	23.1	26.7	24.8	23.6	22.4	18.6	15.2	8.8	(32)	
	10%	WB	1.8	2.9	5.0	6.6	9.5	12.7	15.7	15.5	13.4	9.3	5.2	1.9	(33)	
		MCDB	6.7	8.8	14.2	18.0	22.0	26.5	24.3	23.1	21.8	17.9	13.5	7.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.3	15.5	17.7	18.3	18.9	20.7	17.0	16.5	18.2	18.5	18.3	16.1	(35)	
		MCDBR	20.1	20.9	22.8	22.8	22.5	23.0	19.9	19.9	21.2	22.4	23.2	20.2	(36)	
	5% WB	MCWBR	12.4	11.8	11.8	11.3	10.1	9.5	7.0	6.9	8.6	11.1	12.9	12.5	(37)	
		MCWBR	16.9	17.7	20.3	20.6	20.4	20.0	15.0	14.5	15.8	17.6	19.8	17.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.234	0.242	0.261	0.277	0.288	0.299	0.343	0.336	0.295	0.278	0.243	0.237	(40)	
		taud	2.645	2.627	2.567	2.506	2.518	2.476	2.439	2.423	2.547	2.555	2.623	2.634	(41)	
	Ebn at Noon		1001	1024	1020	1008	993	977	932	935	967	962	978	976	(42)	
		Edh at Noon	69	80	93	105	105	110	113	113	94	85	70	66	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.12	4.06	5.55	6.87	7.74	8.28	7.06	6.46	5.79	4.64	3.52	2.74	(44)	
	RadStd		0.28	0.35	0.35	0.35	0.52	0.37	0.38	0.25	0.20	0.36	0.21	0.19	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page